Your first model in Python

Optimization · Level 2 · Lesson 4



Today we will

- Write the garden model in Python with OR-Tools.
- Read what the solver says: OPTIMAL, FEASIBLE, INFEASIBLE.
- Check the solver’s answer and find a missing rule.

Remember

- Model, variables, constraints, goal, then solve and read the values.
- The solver obeys only the rules you write: check every answer.
- The same program runs on the real OR-Tools in Colab.

Model it together

1. Warm-up (5 min): Too big for graph paper?
2. Rules (10 min): Our problem in plain words.
3. Code (10 min): Type the model together.
4. Run and check (10 min): Is the answer right? Is it fair?
5. Wrap-up (5 min): The rule we forgot.

Practise the model once before class. Python downloads the first time (about 0.3 MB): run it once while online. No phone? Pupils write the model on paper, and you run it later.

Let’s discuss

1. Which line of the program matches which rule on the board?
2. What did the solver get “right” that was still wrong for people?
3. What does INFEASIBLE tell us, and who decides what changes?

Exit ticket

- Write one constraint as a model.add(...) line, and say which rule it stands for.